EXERCISE-I (Conceptual Questions) — Build Up Your Understanding

OCTET RULE :

1. Which among the following shows the limitation of Lewis octet rule ?

(1) CH_4 (2) NO (3) CO_2 (4) NH_4^+

CB0326

2. Which of the following is an example of expanded octet ?

(1) SF_6 (2) PF_5

(3) H_2SO_4 (4) All of these

CB0327

COVALENT BOND :

3. The strength of bonds by 2s - 2s, 2p - 2p and 2p -2s overlapping has the order :-

(1) s – s > p – p > s – p (2) s – s > p – s > p – p

(3) p – p > s – p > s – s (4) p – p > s – s > p – s

CB0003

4. In which of the excitation state of chlorine ClF_3 is formed:-

(1) In ground state

(2) In third excitation state

(3) In first excitation state

(4) In second excitation state

CB0004

5. A sigma bond is formed by the overlapping of :–

(1) s-s orbital alone

(2) s and p orbitals alone

(3) s–s, s–p or p–p orbitals along internuclear axis

(4) p–p orbital along the sides

CB0005

6. Which overlapping is involved in HCl molecule :–

(1) s–s overlap (2) p–p overlap

(3) s–d overlap (4) s–p overlap

CB0006

7. Which is not characteristic of π-bond:-

(1) π - bond is formed when a sigma bond already formed

(2) π - bond are formed from hybrid orbitals

(3) π - bond may be formed by the overlapping of p-orbitals

(4) π-bond results from lateral overlap of atomic orbitals

CB0007

8. π bond is formed :-

(1) By overlapping of hybridised orbitals

(2) Overlapping of s - s orbitals

(3) Head on overlapping of p -p orbitals

(4) By p - p collateral overlapping

CB0008

9. p–p overlapping will be observed in the molecules of:

(1) Hydrogen (2) Hydrogen bromide

(3) Hydrogen chloride (4) Chlorine

CB0010

10. Which compound of xenon is not possible

(1) XeF_2 (2) XeF_4 (3) XeF_5 (4) XeF_6

CB0011

11. Higher is the bond order, greater is -

(1) Bond dissociation energy

(2) Covalent character

(3) Bond length

(4) Paramagnetism

CB0013

12. Which condition is not favourable for the combination of atomic orbitals :-

(1) The combining atomic orbitals nearly have the same energy

(2) The combining atomic orbitals must have the same symmetry about the molecular axis

(3) The combining orbitals must overlap to the maximum extent

(4) The combining orbital must overlap to the minimum extent

CB0328

HYBRIDISATION

13. In the protonation of H_2O, change occurs in

(1) Hybridisation state of oxygen

(2) Shape of molecule

(3) Hybridisation and shape both

(4) None

CB0014

14. In the compound

$\overset{1}{CH_2}=\overset{2}{CH}-\overset{3}{CH_2}-\overset{4}{CH_2}-\overset{5}{C}\equiv\overset{6}{CH}$, the C^2-C^3 bond is formed by the overlapping of :-

(1) $sp - sp^2$ (2) $sp^3 - sp^3$

(3) $sp - sp^3$ (4) $sp^2 - sp^3$

CB0015

ALLEN Session : 2020-21

15. Which of the following elements can not exhibit sp^3d hybridisation state:-
(a) C (b) P (c) Cl (d) B
Correct answer is:-
(1) a, c (2) a, d (3) b, c (4) b, d
CB0018

16. Which of the following species are expected to be planar:-
(a) NH_3 (b) NH_3^{2+} (c) CH_3^+ (d) PCl_3
The correct answer is:-
(1) b and c (2) c and d
(3) b and d (4) a and d
CB0019

17. In which following set of compound/ion has linear shape
(1) CH_4, NH_4^+, BH_4^- (2) CO_3^{-2}, NO_3^-, BF_3
(3) NO_2^+, CO_2, XeF_2 (4) $BeCl_2$, BCl_3, CH_4
CB0020

18. Which of the following set is not correct:–
(1) SO_3, O_3, NH_4^+ all have coordinate bonds
(2) H_2O, NO_2, ClO_2^-, all are 'V' shape molecules
(3) I_3^-, ICl_2^-, NO_2^+ ; all are linear molecules
(4) SF_4, SiF_4, XeF_4 are tetrahedral in shape
CB0022

19. Shape of a molecule having 4 bond pairs and two lone pairs of electrons, will be :-
(1) Square planar (2) Tetra hedral
(3) Linear (4) Octa hedral
CB0023

20. The shape of IF_4^+ will be :-
(1) Square planar
(2) Tetrahedral
(3) Pentagonal bipyramidal
(4) Distorted tetrahedral
CB0024

21. Which of the following has pyramidal shape ?
(1) BF_3 (2) H_3O^+
(3) NO_3^- (4) CO_3^{2-}
CB0025

22. A σ bonded molecule MX_3 is T–shaped. The number of non–bonding pairs of electrons is
(1) 0
(2) 2
(3) 1
(4) Can be predicted if atomic number of M is known.
CB0026

23. Amongst CO_3^{2-}, AsO_3^{3-}, XeO_3, ClO_3^-, BO_3^{3-} and SO_3^{2-} the non-planar species are :-
(1) XeO_3, ClO_3^-, SO_3^{2-}, AsO_3^{3-}
(2) AsO_3^{3-}, XeO_3, CO_3^{-2}
(3) BO_3^{3-}, CO_3^{2-}, SO_3^{2-}
(4) AsO_3^{3-}, BO_3^{3-}, CO_3^{2-}
CB0027

24. The type of hybrid orbitals used by chlorine atom in ClO^-, ClO_2^-, ClO_3^- and ClO_4^- is/are :-
(1) sp, sp^2, sp^3 and sp^3d (2) sp and sp^3
(3) Only sp^3 (4) only sp
CB0028

25. On the basis of hybridization of one s & one p orbitals they are arrange at :-
(1) Two orbitals mutually at 90° angle
(2) two orbitals mutually at 180° angle
(3) Two orbitals mutually at 120°angle
(4) Two orbitals mutually at 150° angle
CB0029

26. Which of the following having a square planar structure is
(1) NH_4^+ (2) BF_4^-
(3) XeF_4 (4) CCl_4
CB0030

27. When p-character of hybridised orbital (formed by s and p orbitals) increases. Then the bond angle
(1) Decreases (2) Increases
(3) Becomes twice (4) Remains unaltered
CB0031

28. Which orbitals overlap to form bond in OF_2
(1) sp^3 – 2p (2) sp^2–2p
(3) sp – 2p (4) p – p
CB0032

ALLEN Session : 2020-21

29. Among the following orbitals/bonds, the angle is minimum between :
(1) sp^3 bonds
(2) p_x and p_y orbitals
(3) H—O—H bond in water
(4) sp bonds

CB0034

30. The AsF_5 molecule is trigonal bipyramidal. The hybrid orbitals used by the As atoms for bonding are:

(1) $d_{x^2-y^2}, d_{z^2}, s, p_x, p_y$ (2) d_{xy}, s, p_x, p_y, p_z

(3) $s, p_x, p_y, p_z, d_{z^2}$ (4) $d_{x^2-y^2}, s, p_x, p_y$

CB0036

31. When the hybridization state of carbon atom changes from sp^3, sp^2 and sp, the angle between the hybridized orbitals.
(1) decrease considerably
(2) increase progressively
(3) decrease gradually
(4) all of these

CB0037

32. The hybridization states of the central atoms of the ions I_3^-, ICl_4^- and ICl_2^- are respectively :
(1) sp^2, sp^3d, sp^3
(2) sp^3d, sp^3d^2 and sp^3d
(3) sp^3d, sp^3d, sp
(4) sp, sp, sp^2

CB0038

33. Molecular shapes of SF_4, CF_4 and XeF_4 are :-
(1) The same, with 2, 0 and 1 lone pairs of electrons respectively
(2) The same, with 1,1 and 1 lone pairs of electrons respectively
(3) Different, with 0,1 and 2 lone pairs of electrons respectively
(4) Different, with 1,0 and 2 lone pairs of electrons respectively

CB0039

34. Which of the following two are isostructural :-
(1) XeF_2, IF_2^- (2) NH_3,BF_3
(3) CO_3^{2-} , SO_3^{2-} (4) PCl_5, ICl_5

CB0040

35. Select the correct matching :

List I	List II
A : XeF_4	1. Pyramidal
B : XeF_6	2. T-shape
C : XeO_3	3. Distorted octahedral
D : $XeOF_2$	4. Square planar

	A	B	C	D
(1)	4	3	1	2
(2)	1	2	3	4
(3)	2	1	3	4
(4)	4	1	3	2

CB0041

36. Which one of the following pair is a correct with respect to molecular formula of xenon compound and hybridization state of xenon in it :
(1) XeF_4 ,sp^3 (2) XeF_2 ,sp
(3) XeF_2 ,sp^3d (4) XeF_4 ,sp^2

CB0042

37. The molecule does not have bent shape :-
(1) SO_2 (2) O_3
(3) H_2O (4) NH_4^+

CB0329

38. Which among the given choices does not have the same hybridisation and geometry of $(PCl_6)^-$?
(1) $(SiF_6)^{2-}$
(2) XeF_6
(3) SF_6
(4) $[Al(H_2O)_6]^{3+}$

CB0330

39. Incorrect regarding the hybridisation is :-
(1) The number of hybrid orbitals are equal to the number of the atomic orbitals that get hybridised
(2) The hybrid orbital are always equivalent in energy and shape
(3) The hybrid orbitals are more effective in forming stable bonds than the pure atomic orbitals
(4) Overlapping and hybridisation are same phenomenon

CB0331

40. The percent s-character in CH_4 is :-
(1) 100% (2) 45% (3) 75% (4) 25%

CB0332

ALLEN Session : 2020-21

DIPOLE MOMENT

41. Which statement is correct:-
(1) All the compounds having polar bonds, have dipole moment
(2) SO_2 is non-polar
(3) H_2O molecule is non polar, having polar bonds
(4) PH_3 is polar molecule having non polar bonds
CB0044

42. BeF_2 has zero dipole moment where as H_2O has a dipole moment because :-
(1) Water is linear
(2) H_2O is bent
(3) F is more electronegative than O
(4) Hydrogen bonding is present in H_2O
CB0045

43. Which of the following molecule have zero dipole moment:-
(1) BF_3 (2) CH_2Cl_2
(3) NF_3 (4) SO_2
CB0047

44. The dipole moment of NH_3 is:-
(1) Less than dipole moment of NCl_3
(2) Higher than dipole moment of NCl_3
(3) Equal to the dipole moment of NCl_3
(4) None of these
CB0048

45. Which set of molecules is polar :-
(1) XeF_4, IF_7, SO_3 (2) PCl_5, C_6H_6, SF_6
(3) $SnCl_2$, SO_2, NO_2 (4) CO_2, CS_2, C_2H_6
CB0050

46. Which of the following has symmetrical structure :
(1) PCl_3 (2) CH_2Cl_2
(3) $CHCl_3$ (4) CCl_4
CB0051

47. Species having zero dipole moment :-
(1) XeF_4
(2) 1,2,4 trichloro benzene
(3) SF_4
(4) CH_2Cl_2
CB0052

48. What conclusion can be drawn from the fact that BF_3 has no dipole moment but PF_3 does
(1) BF_3 is not symmetrical but PF_3 is
(2) BF_3 molecule must be linear
(3) Atomic radius of P is larger than that of B
(4) BF_3 molecule must be planar triangular
CB0053

49. PCl_5 is non polar because :-
(1) P – Cl bond is non-polar
(2) Its dipole moment is zero
(3) P – Cl bond is polar
(4) P & Cl have equal electronegativity
CB0054

50. Dipole moment of CO_2 is zero which implies that :
(1) Carbon and oxygen have equal electronegativities
(2) Carbon has no polar bond
(3) CO_2 is a linear molecule
(4) Carbon has bond moments of zero value
CB0055

51. The correct order of dipole moment is :
(1) $CH_4 < NF_3 < NH_3 < H_2O$
(2) $NF_3 < CH_4 < NH_3 < H_2O$
(3) $NH_3 < NF_3 < CH_4 < H_2O$
(4) $H_2O < NH_3 < NF_3 < CH_4$
CB0056

52. Which of the following has the highest value of dipole moment :
(1) HCl (2) HF (3) HI (4) HBr
CB0058

53. Which compounds has permanent dipole moment:-
(A) BF_3 (B) SiF_4 (C) SF_4 (D) XeF_4
(E) XeF_2 (F) $CHCl_3$
(1) A & B (2) C & D (3) D, E & F (4) C & F
CB0333

54. Which molecule has largest dipole moment :-
(1) CH_4 (2) $CHCl_3$ (3) CCl_4 (4) CHI_3
CB0334

55. Which of the following is the most polar :-
(1) CCl_4 (2) $CHCl_3$ (3) CH_2Cl_2 (4) CH_3Cl
CB0335

56. The molecule does not have zero dipole moment :-
(1) CO_2 (2) CCl_4 (3) BF_3 (4) HCl
CB0336

57. Which of the following compound possess dipole moment :-
(1) Water (2) Boron trifluoride
(3) Benzene (4) Carbon tetra chloride
CB0337

ALLEN Session : 2020-21

MOLECULAR ORBITAL THEORY

58. The ion that is isoelectronic with CO and having same bond order is :-
(1) CN^- (2) O_2^+ (3) O_2^- (4) N_2^+
CB0059

59. Which of the following is paramagnetic:-
(1) O_2^- (2) CN^- (3) CO (4) NO^+
CB0060

60. In the following which of the two are paramagnetic
(a) N_2 (b) CO (c) B_2 (d) NO_2
Correct answer is :-
(1) a and c (2) b and c
(3) c and d (4) b and d
CB0061

61. The bond order of CO molecule on the basis of molecular orbital theory is
(1) Zero (2) 2 (3) 3 (4) 1
CB0062

62. The no. of antibonding electron pair in O_2^- is
(1) 4 (2) 3 (3) 8 (4) 10
CB0064

63. Which of the following species will have the minimum bond energy
(1) N_2 (2) N_2^- (3) N_2^+ (4) N_2^{-2}
CB0065

64. Which of the following ion do not have bond order of 2.5 ?
(1) O_2^- (2) O_2^+ (3) N_2^+ (4) N_2^-
CB0066

65. The electron adds to which one of the following orbitals during change of N_2 to N_2^-.
(1) σ orbital (2) σ* orbital
(3) π* orbital (4) π orbital
CB0338

66. In which of the following species removal of electron exhibit a decrease in paramagentic behavior ?
(1) F_2 (2) N_2 (3) C_2 (4) B_2
CB0339

67. Pick the wrong statement w.r.t. molecular orbital theory (MOT).
(1) Participation of orbitals of comparable energies takes place.
(2) The electron is under the influence of two or more than two nuclie.
(3) The molecular orbital formed is equal to the number of atomic orbitals combined.
(4) The probablity of electron distribution in a molecule is given by atomic orbitals only.
CB0340

68. Which among the following species is does not exist?
(1) He_2^+ (2) H_2^+ (3) H_2^- (4) Be_2
CB0341

69. Which of the given statements is not correct ?
(1) Electron density in a bonding molecular orbital is located between the bonded atoms.
(2) Electron cloud is located away from the space between the nuclei in anti-bonding molecular orbital
(3) In anti-bonding molecular orbital there is always a nodal plane
(4) The total energy of the molecular orbitals varies from the original energies of atomic orbitals
CB0342

70. The given species not having the same bond order as of other three :-
(1) He_2^+ (2) H_2^+ (3) H_2^- (4) Li_2
CB0343

71. The minimum bond length will be observed in :-
(1) O_2 (2) O_2^- (3) O_2^+ (4) O_2^{2-}
CB0344

72. Which among the given pair of molecules consists only of Pi(π) bonds ?
(1) B_2 and F_2 (2) B_2 and C_2
(3) N_2 and O_2 (4) C_2 and N_2
CB0345

73. Which molecule does not exist :-
(1) He_2 (2) O_2 (3) N_2 (4) B_2
CB0346

74. Which is correct electronic configuration for C_2 molecule according to M.O.T.
(1) $KK(\sigma 2s)^2 (\sigma^* 2s)^2 (\pi 2Px^2 = \pi 2Py^2)$
(2) $KK(\sigma 2s^2) (\sigma^* 2s)^2 (\pi 2Px^1 = \pi 2Py^1) \sigma 2Pz^2$
(3) $KK(\sigma^* 2s^2) (\sigma 2s)^2 (\pi 2Px^2 = \pi 2Py^2)$
(4) $KK(\sigma 2s^2) (\sigma^* 2s^2)^2 \sigma 2Pz^2 (\pi 2Px^1 = \pi 2Py^1)$
CB0347

75. The molecule/species having highest bond order :-
(1) O_2 (2) O_2^- (3) O_2^+ (4) O_2^{-2}
CB0348

76. The calculated bond order in H_2^- ion is :-
(1) 0 (2) $\frac{1}{2}$ (3) $-\frac{1}{2}$ (4) 1
CB0349

ALLEN Session : 2020-21

77. O_2 molecule is paramagnetic due to :-

(1) it contains 2 unpaired electrons in π^*2Px and π^*2Py molecular orbitals

(2) it contains no unpaired electrons in π^*2Px and π^*2Py molecular orbitals

(3) it contains 2 paired electrons in $\sigma 2s$ orbital

(4) it contains 1 unpaired electrons in $\sigma 2s$ orbital

CB0350

78. Which is not correct according to M.O.T.

(1) Nb > Na Positive bond order

(2) Nb < Na Negative bond order

(3) Nb = Na Zero bond order

(4) Nb > Na Negative bond order

CB0351

79. The molecule having bond order 3 is :-

(1) H_2 (2) N_2 (3) O_2 (4) He_2^+

CB0352

80. Which is correct electronic configuration for singly positive nitrogen molecule :-

(1) $\sigma 1s^2\ \sigma^*1s^2\ \sigma 2s^2\ \sigma^*2s^2\ \pi 2p_x^2\ \pi 2p_y^2 \sigma 2p_z^1$

(2) $\sigma 1s^2\ \sigma^*1s^2\ \sigma 2s^2\ \sigma^*2s^2\ \sigma 2p_x^2\ \pi 2p_x^2\ \pi^*2p_z^1$

(3) $\sigma 1s^2\ \sigma^*1s^2\ \sigma 2s^2\ \sigma^*2s^2\ \sigma 2p_z^2\ \pi 2p_x^2 \pi 2p_y^2$

(4) $\sigma 1s^2\ \sigma^*1s^2\ \sigma 2s^2\ \sigma^*2s^2\ \sigma 2p_z^2\ \pi 2p_x^1 \pi 2p_z^1$

CB0353

81. Two nodal planes are present in :-

(1) π^*2p_x (2) $\sigma 2p_z$ (3) $\pi 2p_x$ (4) $\pi 2p_y$

CB0354

82. Which is the most stable :-

(1) N_2 (2) N_2^+ (3) N_2^- (4) N_2^{-2}

CB0355

83. Which set of molecules having same sequence of energy levels.

(1) B_2, O_2, N_2 (2) O_2, Be_2, F_2

(3) B_2, C_2, N_2 (4) N_2, O_2, B_2

CB0356

COORDINATE BOND

84. In Co-ordinate bond, the acceptor atoms must essentially contain in its valency shell an orbital:-

(1) With paired electron (2) With single electron

(3) With no electron (4) With three electron

CB0068

85. The bonds present in N_2O_5 are :–

(1) Only ionic

(2) Covalent & coordinate

(3) Only covalent

(4) Covalent & ionic

CB0069

86. Dative bond is present in

(1) SO_3 (2) NH_3

(3) K_2CO_3 (4) BF_3

CB0070

87. The compound containing co-ordinate bond is :

(1) H_2SO_4 (2) O_3

(3) SO_3 (4) All of these

CB0072

WEAK FORCES

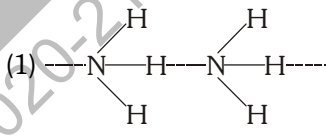
88. Intermolecular hydrogen bonds are not present in:-

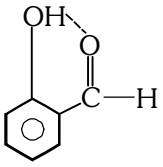
(1) CH_3CH_2OH (2) CH_3COOH

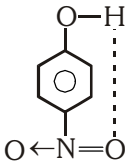
(3) $C_2H_5NH_2$ (4) CH_3OCH_3

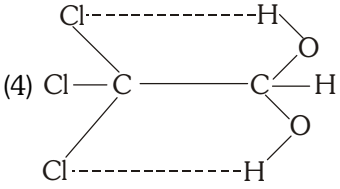
CB0074

89. In which of the following molecule, the shown hydrogen bond is not possible:-

(1) 

(2) 

(3) 

(4) 

CB0075

90. Correct order of volatility is:-

(1) HF > HCl > HBr > HI

(2) HCl > HBr > HI > HF

(3) HI > HBr > HCl > HF

(4) HBr < HCl < HI < HF

CB0076

ALLEN Session : 2020-21

91. The correct order of volatility is:-
(1) $NH_3 < H_2O$
(2) p– nitro phenol < o– nitro phenol
(3) $CH_3OH > CH_3 – O – CH_3$
(4) HF > HCl

CB0077

92. The incorrect order of decreasing boiling points is
(1) $NH_3 > AsH_3 > PH_3$
(2) $H_2O > H_2Se > H_2S$
(3) $Br_2 > Cl_2 > F_2$
(4) $CH_4 > GeH_4 > SiH_4$

CB0078

93. Acetic acid exists as dimer in benzene due to:-
(1) Condensation reaction
(2) Hydrogen bonding
(3) Presence of carboxyl group
(4) None of the above

CB0079

94. Maximum no. of hydrogen bonds formed by a water molecule in ice is
(1) 4 (2) 3 (3) 2 (4) 1

CB0080

95. Strongest hydrogen bond is shown by :
(1) Water (2) Ammonia
(3) Hydrogen fluoride (4) Hydrogen sulphide

CB0081

96. Density of ice is less than that of water because of
(1) presence hydrogen bonding
(2) crystal modification of ice
(3) open porous structure of ice due to hydrogen bonding
(4) different physical states of these

CB0082

97. NH_3 has abnormally high boiling point because it has :
(1) Alkaline nature (2) Distorted shape
(3) sp^3 - Hybridisation (4) Hydrogen bonding

CB0083

98. Which of the following is soluble in water ?
(1) CS_2 (2) C_2H_5OH
(3) CCl_4 (4) $CHCl_3$

CB0084

99. KF combines with HF to form KHF_2. The compound contains the species :
(1) K^+, F^- and H^+ (2) K^+, F^- and HF
(3) K^+ and $[HF_2]^-$ (4) $[KHF]^+$ and F_2

CB0085

IONIC BOND

100. Which one is the correct statement with reference to solubility of $MgSO_4$ in water:
(1) Hydration energy of $MgSO_4$ is higher in comparison to its lattice energy
(2) Ionic potential of Mg^{2+} is very low
(3) SO_4^{2-} ion mainly contributes towards hydration energy
(4) Size of Mg^{2+} and SO_4^{2-} are similar

CB0088

101. Conditions for ionic bond formation is/are :
(a) Small cation, large anion
(b) Low IP of cation, high electron affinity of anion
(c) Large cation, small anion and less charge
(d) Less lattice enrgy
Correct answer is:
(1) a, d (2) b, c and d
(3) b and c (4) a, b

CB0089

102. The force responsible for dissolution of ionic compound in water is –
(1) Dipole – dipole forces
(2) Ion – dipole force
(3) Ion – ion force
(4) Hydrogen bond

CB0091

103. Born Haber cycle is mainly used to determine
(1) Lattice energy (2) Electron affinity
(3) Ionisation energy (4) Electronegativity

CB0092

104. An ionic compound A^+B^- is most likely to be formed when –
(1) Ionization energy of A is low
(2) Electron affinity of B is high
(3) Electron affinity of B is low
(4) Both (1) and (2)

CB0093

105. Highest melting point would be of
(1) $AlCl_3$ (2) LiCl
(3) NaCl (4) $BeCl_2$

CB0096

106. As compared to covalent compounds electrovalent compounds generally possess
(1) High m.p. and high b.p.
(2) Low m.p. and low b.p.
(3) Low m.p. and high b.p.
(4) high m.p. and low b.p.

CB0098

ALLEN Session : 2020-21

107. The electronic configuration of metal M is $1s^2\ 2s^2\ 2p^6\ 3s^1$. The formula of its oxide will be :
(1) MO (2) M_2O
(3) M_2O_3 (4) MO_2
CB0099

108. Which of the following does not show electrical conduction ?
(1) diamond
(2) graphite
(3) sodium chloride (fused)
(4) potassium
CB0102

109. The most covalent halide is:-
(1) AlF_3 (2) $AlCl_3$
(3) $AlBr_3$ (4) AlI_3
CB0103

110. Ionic potential (ϕ) of electropositive element will be highest in which of the following compound:-
(1) CsCl (2) $MgCl_2$
(3) AlF_3 (4) SF_6
CB0104

111. LiCl is soluble in organic solvent while NaCl is not because :-
(1) Lattice energy of NaCl is less than that of LiCl
(2) Ionisation potential of Li is more than that of Na
(3) Li^+ has more hydration energy than Na^+ ion
(4) LiCl is more covalent compound than that NaCl
CB0105

112. The most stable carbonate is
(1) Li_2CO_3 (2) $BeCO_3$
(3) $CaCO_3$ (4) $BaCO_3$
CB0106

113. Correct order of covalent character of alkaline earth metal chloride in
(1) $BeCl_2 < MgCl_2 < CaCl_2 < SrCl_2$
(2) $BeCl_2 < CaCl_2 < SrCl_2 < MgCl_2$
(3) $BeCl_2 > MgCl_2 > CaCl_2 > SrCl_2$
(4) $SrCl_2 > BeCl_2 > CaCl_2 > Mg\ Cl_2$
CB0107

114. Which of the compound is least soluble in water
(1) AgF (2) AgCl
(3) AgBr (4) AgI
CB0108

115. CCl_4 is more covalent than LiCl because :
(1) There is more polarization of Cl in CCl_4
(2) There is more polarization of Cl in LiCl
(3) CCl_4 has more weight
(4) None of above
CB0109

116. The correct order of decreasing polarisable ions is:
(1) Cl^-, Br^-, I^-, F^- (2) F^-, I^-, Br^-, Cl^-
(3) F^-, Cl^-, Br^-, I^- (4) I^-, Br^-, Cl^-, F^-
CB0111

117. Ionic conductances of hydrated M^+ ions are in the order –
(1) $Li^+(aq) > Na^+(aq) > K^+(aq) > Rb^+(aq) > Cs^+(aq)$
(2) $Li^+(aq) > Na^+(aq) < K^+(aq) < Rb^+(aq) < Cs^+(aq)$
(3) $Li^+(aq) > Na^+(aq) > K^+(aq) > Rb^+(aq) < Cs^+(aq)$
(4) $Li^+(aq) < Na^+(aq) < K^+(aq) < Rb^+(aq) < Cs^+(aq)$
CB0112

118. Which of the following does not give an oxide on heating –
(1) $MgCO_3$ (2) Li_2CO_3
(3) $ZnCO_3$ (4) K_2CO_3
CB0114

119. Which decomposes on heating –
(1) NaOH (2) KOH
(3) LiOH (4) RbOH
CB0115

120. Which of the following forms metal oxide on heating
(1) Na_2CO_3 (2) Li_2CO_3
(3) K_2SO_4 (4) $NaHCO_3$
CB0116

121. Increasing order of stability of –
I. K_2CO_3 II. $MgCO_3$ III Na_2CO_3
(1) I < II < III (2) II < III < I
(3) II < I < III (4) I < III < II
CB0117

122. Which of the following carbonate will not decompose on heating :-
(1) $BaCO_3$ (2) $ZnCO_3$
(3) Na_2CO_3 (4) Li_2CO_3
CB0118

123. Correct order of melting point is ?
(1) $SnCl_2 > SnCl_4$ (2) $SnCl_4 > SnCl_2$
(3) $SnCl_2 = SnCl_4$ (4) None of these
CB00357

ALLEN Session : 2020-21

124. Which of the following suffers a weight loss on heating :-

(1) Li_2CO_3 (2) Washing soda

(3) both (1) & (2) (4) None

CB0358

125. On heating Na_2CO_3 gives :-

(1) $Na_2O + CO_2$ (2) $Na_2 + CO_3$

(3) $Na + CO_2$ (4) None

CB0359

126. Correct order of melting point is :-

(1) $NaF < MgF_2 < AlF_3$ (2) $AlF_3 > NaF > MgF_2$

(3) $MgF_2 < NaF < AlF_3$ (4) None

CB0360

127. Which of the following option is incorrect regarding electrovalent bond ?

(1) Ionic bond will be formed more easily between elements with comparatively low ionisation enthalpy & high negative electron gain enthalpy

(2) NH_4^+ is an exception of cation : because it made up of two non metallic element

(3) Lattice enthalpy plays a key role in the formation of ionic compound

(4) All of these

CB0361

ALLEN Session : 2020-21

EXERCISE-I (Conceptual Questions) ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**2**	**4**	**3**	**3**	**3**	**4**	**2**	**4**	**4**	**3**	**1**	**4**	**2**	**4**	**2**
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	**1**	**3**	**4**	**1**	**4**	**2**	**2**	**1**	**3**	**2**	**3**	**1**	**1**	**2**	**3**
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	**2**	**2**	**4**	**1**	**1**	**3**	**4**	**2**	**4**	**4**	**4**	**2**	**1**	**2**	**3**
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	**4**	**1**	**3**	**2**	**3**	**1**	**2**	**4**	**2**	**4**	**4**	**1**	**1**	**1**	**3**
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	**3**	**2**	**4**	**1**	**3**	**4**	**4**	**4**	**4**	**4**	**3**	**2**	**1**	**1**	**3**
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	**2**	**1**	**4**	**2**	**1**	**1**	**1**	**3**	**3**	**2**	**1**	**4**	**1**	**3**	**2**
Que.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Ans.	**2**	**4**	**2**	**1**	**3**	**3**	**4**	**2**	**3**	**1**	**3**	**2**	**1**	**4**	**3**
Que.	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Ans.	**1**	**2**	**1**	**4**	**4**	**4**	**4**	**3**	**4**	**1**	**4**	**4**	**4**	**3**	**2**
Que.	121	122	123	124	125	126	127								
Ans.	**2**	**3**	**1**	**2**	**4**	**1**	**4**								

EXERCISE-III (Analytical Questions) Master Your Understanding

1. The thermal stability of alkaline earth metal carbonates $MgCO_3$, $CaCO_3$, $BaCO_3$ and $SrCO_3$ decreases as :
(1) $CaCO_3 > SrCO_3 > MgCO_3 > BrCO_3$
(2) $BaCO_3 > SrCO_3 > MgCO_3 > CaCO_3$
(3) $BaCO_3 > SrCO_3 > CaCO_3 > MgCO_3$
(4) $MgCO_3 > CaCO_3 > SrCO_3 > BaCO_3$
CB0224

2. Which of the following are iosoelectronic and isostructural NO_3^-, CO_3^{-2}, ClO_3^-, SO_3
(1) NO_3^-, CO_3^{-2} (2) SO_3, NO_3^-
(3) ClO_3^-, CO_3^{-2} (4) CO_3^{-2}, SO_3
CB0225

3. According to molecular orbital theory which of the following statement about the magnetic character and bond order is correct regarding O_2^+ ?
(1) paramagnetic and bond order < O_2
(2) paramagnetic and bond order > O_2
(3) diamagnetic and bond order < O_2
(4) diamagntic and bond order > O_2
CB0226

4. Number of σ and π bond in SO_4^{-2} are :–
(1) 4, 2 (2) 4, 3
(3) 4, 4 (4) 2, 3
CB0228

5. The paramagnetic property of the oxygen molecule is due to the presence of unpaired electrons present in :-
(1) $(\sigma 2p_x)^1$ and $(\sigma^* 2p_x)^1$ (2) $(\sigma 2p_x)^1$ and $(\pi 2p_y)^1$
(3) $(\pi^* 2p_y)^1$ and $(\pi^* 2p_x)^1$ (4) $(\pi^* 2p_y)^1$ and $(\pi 2p_y)^1$
CB0229

6. What is the hybridization of oxygen atom in an alcohol molecule?
(1) sp^3 (2) sp (3) sp^2 (4) sp^3d
CB0233

7. In which of the following ionisation processes, the bond order has increased and the magnetic behaviour has changed
(1) $NO \rightarrow NO^+$ (2) $O_2 \rightarrow O_2^+$
(3) $N_2 \rightarrow N_2^+$ (4) $C_2 \rightarrow C_2^+$
CB0235

8. Which of the following hydrogen bonds is the strongest
(1) F–H.....F (2) O–H.....O
(3) O–H.....F (4) O–H.....N
CB0236

9. Which of the following species exhibits the diamagnetic behaviour
(1) O_2^+ (2) O_2 (3) NO (4) O_2^{2-}
CB0237

10. The charge/size ratio of a cation determines its polarizing power. Which one of the following sequences represents the increasing order of the polarizing power of the cationic species, K^+, Ca^{+2}, Mg^{+2}, Be^{+2}
(1) $Be^{+2} < K^+ < Ca^{+2} < Mg^{+2}$
(2) $K^+ < Ca^{+2} < Mg^{+2} < Be^{+2}$
(3) $Ca^{+2} < Mg^{+2} < Be^{+2} < K^+$
(4) $Mg^{+2} < Be^{+2} < K^+ < Ca^{+2}$
CB0238

11. Which one of the following constitutes a group of the isoelectronic species ?
(1) C_2^{2-}, O_2^-, CO, NO
(2) $NO^+, C_2^{2-}, CN^-, N_2$
(3) $CN^-, N_2, O_2^{2-}, C_2^{2-}$
(4) N_2, O_2^-, NO^+, CO
CB0239

12. Which of the following pairs of species have the same bond order ?
(1) CN^- and NO^+ (2) CN^- and CN^+
(3) O_2^- and CN^- (4) NO^+ and CN^+
CB0240

13. Among the following, the molecule with highest dipole moment is :-
(1) CH_3Cl (2) CH_2Cl_2
(3) $CHCl_3$ (4) CCl_4
CB0241

14. Which one of the following molecules is expected to exhibit diamagnetic behaviour ?
(1) C_2 (2) N_2^- (3) O_2 (4) S_2
CB0243

ALLEN Session : 2020-21

15. Which one of the following order is correct for the bond energies of halogen molecules :
(1) $I_2 > Cl_2 > Br_2$ (2) $Br_2 > Cl_2 > I_2$
(3) $I_2 > Br_2 > Cl_2$ (4) $Cl_2 > Br_2 > I_2$

CB0245

16. The boiling point of ICl is nearly 40°C higher than that of Br_2 although the two substances have the same relative molecular mass. This is bacasue :-
(1) ICl is ionic compound
(2) I-Cl bond is stronger than Br - Br bond
(3) ICl is polar covalent molecule while Br_2 is non polar
(4) IP of Iodine is less than that of Br.

CB0246

17. Nitrogen does not form NF_5 because
(1) Nitrogen is member of V group
(2) It contains no empty d-orbital
(3) The bond energy of N ≡ N is very high
(4) Inert pair effect exists in the molecule

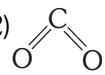
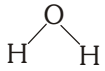
CB0247

18. CO_2 is a gas, while SiO_2 is a solid but both are-
(1) Covalent containing π–bond
(2) Molecules having pπ – dπ bonding
(3) Acidic
(4) Discrete molecules

CB0248

19. Dipole moment is shown by
(1) 1, 4 - dichlorobenzene
(2) Cis 1, 2-dichloro ethene
(3) Trans-1, 2– dichloro ethene
(4) benzene

CB0250

20. Experiment shows that H_2O has a dipole moment where as CO_2 has not. Point out the structures which best illustrate these facts –
(1) O=C=O, H—O—H
(2)  , H—O—H
(3) O=C=O, 
(4) 

CB0251

21. In which of the species, bond order increases on removing one electron:-
(a) NO (b) CN^- (c) O_2 (d) CO
Correct answer is -
(1) b and d (2) a and c
(3) b, d and c (4) b and c

CB0252

22. Increasing order of bond length in NO, NO^+ and NO^- is :-
(1) $NO > NO^- > NO^+$
(2) $NO^+ < NO < NO^-$
(3) $NO < NO^+ < NO^-$
(4) $NO < NO^+ = NO^-$

CB0253

23. N_2 and O_2 are converted into monoanions, N_2^- and O_2^- respectively. Which of the following statements is wrong ?
(1) In N_2^- , N-N bond weakens
(2) In O_2^-, O-O bond order increases
(3) In O_2^-, O-O bond order decreases
(4) N_2^- becomes pagamegnetic

CB0254

24. N_2 and O_2 are converted into monocations, N_2^+ and O_2^+ respectively. Which of the following is wrong ?
(1) In N_2^+, N-N bond weakens
(2) In O_2^+, the O-O bond order increases
(3) In O_2^+, paramagnetism decreases
(4) N_2^+ becomes diamagnetic

CB0255

25. Glycerol is more viscous than glycol the reason is :-
(1) Higher molecular wt.
(2) More covalent
(3) More extent of hydrogen bonding
(4) Complex structure

CB0258

26. Incorrect order of viscosity :-
(1) $H_2SO_4 > HNO_3$
(2) $H_2O > CH_3OH$
(3) o-nitro phenol > p-nitro phenol
(4) Glycol > ether

CB0259

27. Which of the following does not form a hydrogen bond with water
(1) $(CH_3)_2CO$ (2) CH_3CN
(3) CH_3OH (4) C_2H_6

CB0261

ALLEN Session : 2020-21

28. Lithium chloride is highly soluble in –
(1) C_6H_6 (2) H_2O (3) D_2O (4) All
CB0264

29. Which of the following molecule contains net π-bond only
(1) B_2 (2) C_2
(3) C_2^{-2} (4) Both 1 & 2
CB0266

30. Pick out the incorrect statement :-
(1) sp^3d hybridisation involves $d_{x^2-y^2}$ orbital
(2) Hybridised orbital form σ-bond when overlaps with other orbitals.
(3) SF_2 molecule is more polar than CS_2
(4) o-nitrophenol is more volatile than p-nitrophenol.
CB0267

31. The group of substances in which at melting point covalent bond becomes weak :-
(1) NaCl, KCl, $CaCl_2$ (2) $I_{2(S)}$, $CH_{4(S)}$, dry ice
(3) B_4C, dimond, SiC (4) All of the above
CB0268

32. The incorrect statement is :-
(1) π^* p have two nodal planes
(2) Bond order of HeH^+ is 0.5
(3) In NCO^- , C is sp hybridised
(4) O_3 is polar while O_2 is non polar
CB0269

33. Solid CH_4 is
(1) Molecular solid (2) Ionic solid
(3) Covalent solid (4) None of these
CB0270

34. Which compound has the weakest bond
(1) Diamond
(2) Neon(solid)
(3) KCl
(4) water (ice)
CB0271

35. Which of the following order is not correct ?
(1) $SF_2 > SF_4 > SF_6$ (ionic character)
(2) $AlF_3 < Al_2O_3 <$ AlN (covalent character)
(3) $CaCl_2 < SnCl_2 < CdCl_2$ (covalent character)
(4) $ZnCl_2 < CdCl_2 < HgCl_2$ (ionic character)
CB0274

36. Match the column :-

(A) $NO_3^- + HCl$	(p) Keesom attraction
(B) $Xe + H_2O$	(q) Debye attraction
(C) $CH_3-\overset{\overset{O}{\|}}{C}-CH_3+CH_3-C\equiv N$	(r) London force
(D) $CO_2 + CS_2$	(s) Ion-dipole attraction

(1) A-s, B-q, C-p, D-r
(2) A-q, B-s, C-p, D-r
(3) A-s, B-p, C-q, D-r
(4) A-s, B-q, C-r, D-p
CB0275

37. Which of the following statement is not correct ?
(1) NO has one unpaired electron in ABMO.
(2) N–H bond length is higher in N_2H_4 as compared to N_2H_2
(3) Both PF_5 and IF_5 have identical shape
(4) Both SO_2 and BrF_3 are polar and planar
CB0277

38. Which of the following follow octet rule ?
(1) $BeH_{2(s)}$ (2) B_2H_6
(3) $BeCl_{2(s)}$ (4) None
CB0371

39. Total no. of 90° angles in IF_5 molecule ?
(1) 4 (2) 8 (3) 0 (4) 6
CB0372

40. Which is most nonpolar compound :-
(1) PCl_5 (2) PCl_3
(3) CH_3Cl (4) ClF_7
CB0373

41. C_2 molecule consists of both π bonds due to :-
(1) The presence of four electrons in two π molecular orbitals
(2) The absence of any electrons in two π molecular orbitals
(3) The presence of 2 electrons in one σ molecular orbital
(4) The presence of 4 electrons in one σ molecular orbital
CB0374

ALLEN Session : 2020-21

42. Which is incorrect for both $H_2^{\ominus}$ and $He_2^{\oplus}$ ions :-

(1) both have same bond order

(2) both have same number of electrons in antibonding M.O.

(3) both have similar stability

(4) both have different bond order

CB0375

43. Incorrect about PCl_5 molecules is :-

(1) Three P–Cl bond lie in equatorial plane

(2) Two P–Cl bond lie in axial plane

(3) Axial bond pairs suffer more repulsive interaction from the equatorial bond pair

(4) Equatorial bonds are longer than the axial bonds

CB0376

44. Incorrect about C_2 is :-

(1) It contain 2 π bond

(2) It contain $KK(\sigma 2s)^2\,(\sigma^*2s)^2\,(\pi 2P_x^{\,2} = \pi 2P_y^{\,2})$

(3) It contains four electrons in 2 pi molecular orbitals

(4) It contains 6 electrons in bonding molecular orbital

CB0377

ALLEN
Session : 2020-21

EXERCISE-III (Analytical Questions) ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**3**	**1**	**2**	**1**	**3**	**1**	**1**	**1**	**4**	**2**	**2**	**1**	**1**	**1**	**4**
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	**3**	**2**	**3**	**2**	**3**	**2**	**2**	**2**	**4**	**3**	**3**	**4**	**1**	**4**	**1**
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	
Ans.	**3**	**2**	**1**	**2**	**4**	**1**	**3**	**3**	**3**	**1**	**1**	**4**	**4**	**4**	